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Porting a Process-Based Crop Model to a High-Performance Computing Environment for Plant Simulation

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***Abstract*—Increasing concerns about food security have stimulated integrated assessment of the sustainability of agricultural systems at regional, national and global scales with high-resolution. Traditionally, the process-based agricultural models are designed for field scale studies that obtain inputs, run the simulations and provide outputs through the graphic interface. The graphic interface based model dose not suit for modelling practices requiring a large number of simulations. Here, we developed a high performance approach which concurrently executed the Agricultural Production Systems sIMulator (APSIM) simulations using parallel programming techniques. In this approach, an APSIM simulation template with replaceable parameters was firstly designed, and new simulations based on the template was then constructed by dynamically replacing parameters of climate, soil and management options. We parallelized the batched running method in a shared-memory multiprocessor system using Python's Multiprocessing module. We tested the approach with a case study that simulated the productivity of continuous wheat cropping system during 20 years period along the Australian cereal−growing regions under management practices of 5 levels nitrogen application and 3 stubble management practices. More than 170 K runs were finished in 43h by using 64 workers, achieved a speedup ratio of 60. The parallelized method proposed in this study makes large-scale and high-resolution agricultural systems assessment possible.**

Keywords: APSIM; parallel processing; batched running; environmental modelling

I. INTRODUCTION

In the application of crop models, such as sensitivity analysis, model calibration, and integrated agricultural systems assessments over large geographic extent at a high resolution, models need to be executed for a large number of times with different parameters and driving data [1-3]. Many process-based models simulate the crop growth and its interactions with surrounding environment in a daily, even hourly time step, which are computational intensive that limits the extent of these models could be applied [4]. Several studies have been conducted to address the computational challenge through parallel processing and high-performance computing (HPC) technology. For example, In [4], the authors developed an efficient approach to implement high-resolution environmental modelling with Environmental Policy Integrated Climate (EPIC) by utilizing the Portable Batch System, a job queuing and schedule software, and achieved 40 times speedup *via* a UNIX HPC cluster which had 20 nodes [5]. In [6], the author described the potential benefits of parallelizing array-based in-memory model through Graphics Processing Units (GPUs) cluster in modelling social−economic aspects of the global environmental changes. In [7], the authors presented a GPU-enabled 2D flood model that overcame the barrier of compute efficiency for real-time simulation of flood flows. The states of these studies were generally in a pilot phase or specifically designed for a certain computing environments or model.

In this study, we proposed an approach for porting a process-based crop model, e.g. the Agricultural Production Systems sIMulator (APSIM), to a high-performance computing environment. We tested the approach through a case study of modelling agricultural systems over Australian cereal-growing regions under 15 agricultural management practices. We provided the key steps to implement the batched and parallelized running method and discussed the potential opportunities to improve the method with the development of hardware.

II. METHODOLOGY

A. APSIM

APSIM is a widely used modelling framework that simulates crop growth, biomass, yield, soil organic carbon, water use efficiency, nitrogen use efficiency and crop's interaction with soil and atmospheric environment in a daily step under various farming conditions [8-12] . It can simulate a wide variety of crops with a flexible management rule designing environment. It has been systematically validated under a wide range of soil and climate conditions around the world [13-16]. APSIM has traditionally been used for field study that it obtains input data and conducts the simulation through the user interface that lacks the ability for running simulations with a parameters updating mechanism [17], despite the fact that some applications need running a large number of simulations, e.g. parameter optimization and sensitivity analysis, a batched running mode is highly desirable.

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B. Porting APSIM from Interface to Batched Processing

APSIM has two types of simulation files, .apsim and .sim, that both of them are stored in a XML format. The differences between .apsim and .sim were listed in Table. I. The .sim file is a standalone file that includes all parameters for an integrated simulation. To run the simulations in a batched mode, we need dynamically replace the parameters and create new .sim files and execute them. So, we chose the .sim format. We first designed an .apsim format model template through the user interface, which configured all the input and output parameters and management options. We converted .apsim template into .sim file. We developed a Python script which could replace and create .sim files in real time. Soil, climate, and management option parameter are required to be updated for each simulation (Figure 1). The XML file processing was conducted by Python XML module which read the template file, searched the nodes for parameter, replace the parameter and created a new .sim file.

C. Computing Environment

As APSIM is a Windows-based application, a shared-memory multi-processor server supported by Windows Server 2008 R2 was employed in this study. The server had four Intel(R) Xeon(R) X7560 @ CPUs. Each CPU had 8 cores with 2.27 GHz clock speed and 24 Mb L3 cache. The total Random Access Memory (RAM) was 512G. Hyper-threading technology doubled the machine concurrently processing capacity to 64.

D. Parallel Strategies

The batched running method saved a significant time by automatically producing and executing the simulations. It still cannot satisfy applications needs a large number of simulations. So, it is worthwhile to promote the compute efficiency further by parallel processing. Parallel processing could conduct multiple instructions or run many processes concurrently, thereby accelerating the computing efficiency. It suits for applications which are standalone and have no frequent communications with each other. We implemented the parallelism of APSIM simulations with Python Multiprocessing package which

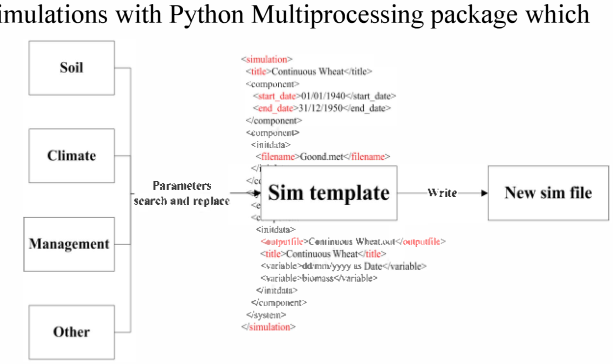

Figure 1 Schematic of APSIM-sim files creation using a dynamic parameter substitution strategy.

realizes parallel processing by spawning the same number of processes as processors in the machine. The workflow of our parallelized implementation was shown in Figure 2. The script first arranged a list of parameters of soil, climate and management regimes. The Multiprocessing module discovered the quantity of processors in the target machine and then created a pool of processes as workers. When the worker processes were spawned, the script allocated the simulation template and parameters to each process. The process applied the aforementioned batched running method to produce the .sim file and used a sub-process to execute the apsim.exe with the produced .sim file as a parameter. The sub-process was created by Python Subprocess module that can execute an application by a command line. The Subprocess module has two functions for running command lines, e.g. Call and Popen. The major difference between them is that the Call function waits the process to be finished but the Popen not. Because the multiprocessing created a process for each core, so we use the function of Call here to avoid spawning too many processes. When the simulations finished, the master process collected the results.

III. CASE STUDY

A. Climate, Soil Data and Management Practices

Our study area covered the Australia's cereal-growing and marginal regions (100km buffer from the cereal-growing regions) (Figure 3). We divided the entire study area into 11,575 climate-soil units (CS units) which had relative similar climate and soil properties. National-scale grid climate data from 1989 to

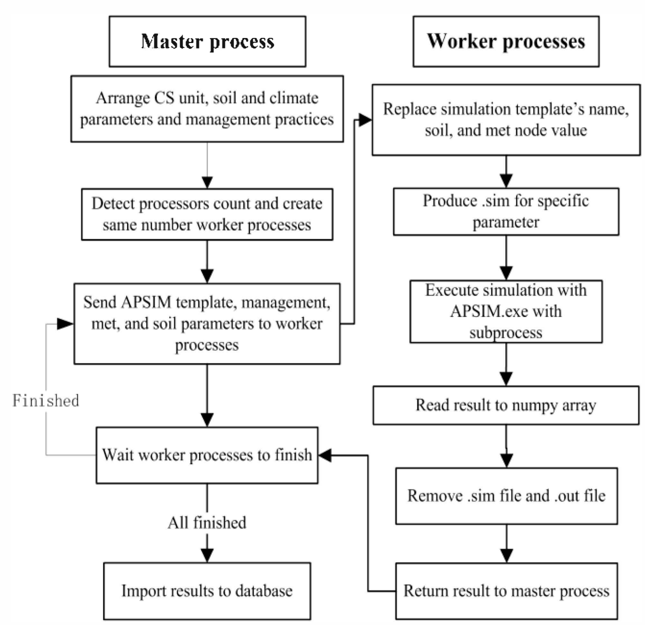

Figure 2. The workflow of parallel processing APSIM simulations with Python Multiprocessing module. Numpy is Python numeric package and Numpy array is a data structure for storing sequence data type.

TABLE I. THE CHARACTERISTICS OF TWO FORMAT OF APSIM PROJECT FILES

	Interface editable	*Stand alone*	*Running application*	*Command line*	*Management strategies*	*Climate parameters*	*Soil parameters*	*Cultivar parameters*	*Model parameters*
.apsim	Yes	No	ApsimRun.exe	Yes	Yes	Yes	Yes	No	No
.sim	No	Yes	Apsim.exe	Yes	Yes	Yes	Yes	Yes	Yes

2009 with a 0.05° spatial resolution were obtained from the Bureau of Meteorology, Australia (http://www.bom.gov.au). The grid datasets were interpolated from roughly 4600 weather stations across the country [18]. The statistics (mean value) of five daily climate variables (maximum and minimum temperature, solar radiation, rainfall, and evaporation) were summarized for each CS unit [19]. Soil data was extracted from Australian Soil Resource Information System (ASRIS) [20]. Soil properties in ASRIS were stored in vector layers. The layers were firstly converted into grid format and then summarized for CS units by Zonal statistics. Soil properties included pH, Bulk Density (BD), Drained Upper Limit (DUL), 15bar Lower Limit (LL15), and Layer Depth (LD). We designed 15 management practices, 5 fertilization levels (e.g. 0, 50, 100, 150, 200 kg N ha^{-1}) and 3 residue removal rates (e.g. %0, 50% and 100%) on crop productivity which resulted in almost 173 K simulations.

B. Compute Time and Speedup Ratio

The computational challenges of this study are mainly sourced from the large number of simulations and the long wall-time of each simulation (54 seconds). In this study, the input climate and soil data were named with respect to the ID of the CS unit. They then combined with the management scenarios and queued into a job list. From this perspective, these jobs can be classified as a typical Single Instruction Multiple Data (SIMD) parallelism. In total over 20 year 173 K simulations were required to be executed to cover the agricultural systems of our study area. It required almost 2580h of computing time on a desktop using serial batch computing method. Using our shared-memory multi-processor system, the simulations were spitted into 64 processes of workers, and the total execution time was 43h. The speedup ratio of the parallelized method was 60 compared with the serial application. Furthermore, the results were imported to relational database for statistic, analysis, and visualization.

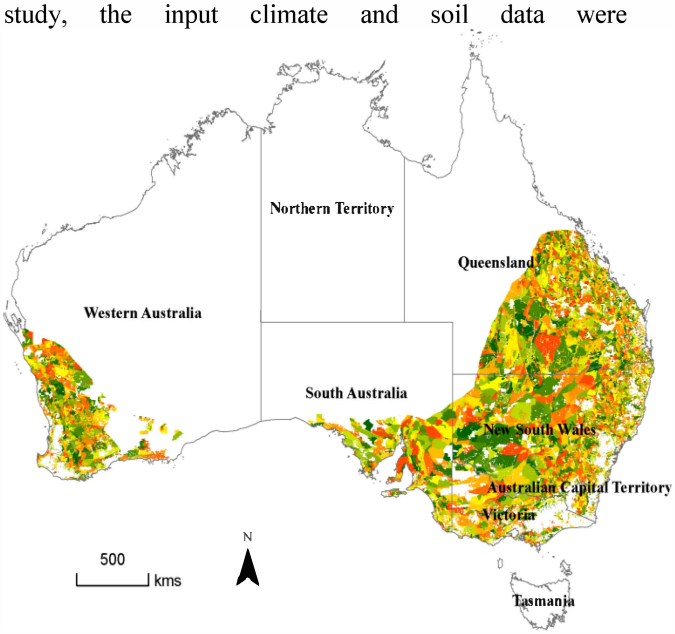

Figure 3. Climate-soil zones. Each zone is demonstrated by one unique colour.

IV. DISCUSSION AND CONCLUSION

Only a small number of studies about running process-based crop model in a HPC environment have been reported. In this study, we developed a method for batched running of APSIM simulations which overcame the low efficiency of interface-based APSIM. We embedded the batched method into Python multiprocessing module for concurrently executing the simulations. The parallel computing overcame the computing efficiency issues faced high-resolution agricultural systems modelling over large geographic extent. This paper described the details of the implementation of the parallel programming which could help porting other crop models to a HPC environment. We finished the hundreds of thousands of APSIM simulations for national scale study with high-resolution in an accepatable time period of 43h. The capability of our method has greatly broadened the extent and resolution could be studied for solving global environmetal change challenges in agricultral aspect.

In addition, APSIM is a complex process-based model that simulates many biophysical processes in a farming system that it cannot be directly translated into a GPU computing environment. At this stage, GPU only suits for simple linear models operating on large datasets that complex models like APSIM are not suitable for implantation in a GPU computing environment. With the development of GPGPU technology, porting APSIM to the GPU computing environment will be a faster and hardware−inexpensive option.

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